SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) **Total Marks: 100**

ANNEXURE A

Assignment 1

Code of Conduct:

I p.sowmya sri(192425265) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P.SOWMYA SRI

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Questions

Annexure A

Assignment 1

Question 30 – Hospital Appointment Scheduler

Problem Statement:

Patients book appointments with doctors.

Questions

a) Develop a Java program using **LinkedList**.

b) Traverse appointments using Iterator.

c) Compare LinkedList and ArrayList.

d) Explain scheduling systems.

A)

HospitalAppointment.java

Share Run

```java
import java.util.*;

public class HospitalAppointment {
    public static void main(String[] args) {
        LinkedList<String> appointments = new LinkedList<>();

        appointments.add("Ravi - Dr. Kumar");
        appointments.add("Priya - Dr. Meena");
        appointments.add("Arun - Dr. Raj");

        Iterator<String> it = appointments.iterator();

        System.out.println("Appointments:");
        while (it.hasNext()) {
            System.out.println(it.next());
        }
    }
}
```

Output

```
Appointments:
Ravi - Dr. Kumar
Priya - Dr. Meena
Arun - Dr. Raj

=== Code Execution Successful ===
```

Questions

b) Traverse Appointments using Iterator

- Create an Iterator using iterator().
- Use hasNext() to check for the next appointment.
- Use next() to display each appointment one by one

c) Compare LinkedList and ArrayList

LinkedList	ArrayList
Uses doubly linked list	Uses dynamic array
Faster insertion/deletion	Faster random access
More memory required	Less memory required
Slower for accessing elements	Faster for accessing elements

d) Explain Scheduling Systems

A hospital scheduling system manages patient appointments with doctors efficiently. It helps avoid double bookings, reduces waiting time, and allows easy appointment booking, cancellation, and rescheduling. It improves hospital workflow and patient satisfaction.